Milestone 3 | Detection results

IE 7615 - Group 1 | Evaluation and report: Yosephine Tong

We tested Aditi's saved YOLOv8n model on the held-out Milestone 2 dataset: 36 images containing 113 labelled faces.

Metric	Test result
mAP@0.5	96.2%
mAP@0.5:0.95	86.2%
Precision	89.8%
Recall	93.8%
Mean IoU of correct matches	0.931

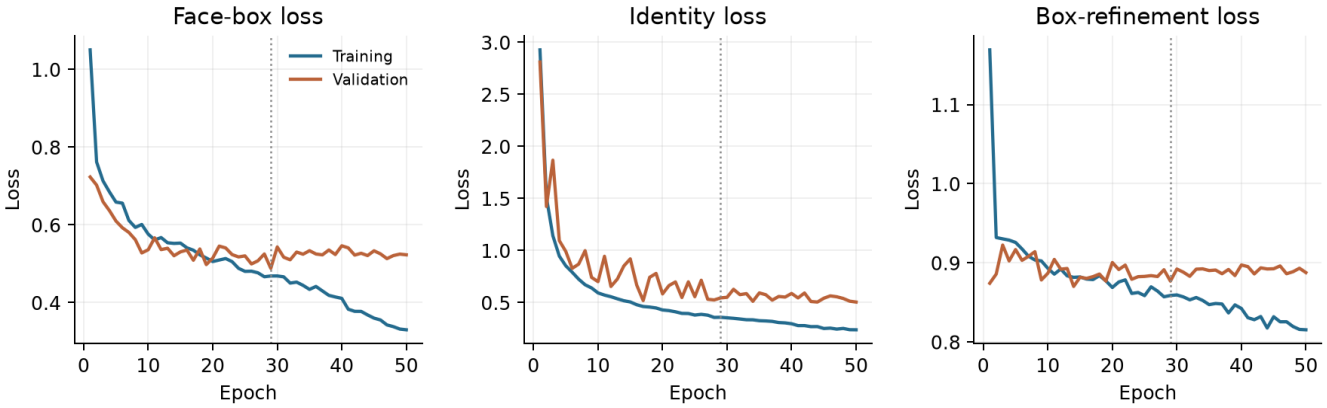
Precision and recall use confidence 0.25. A correct match needs the right identity and box overlap (IoU) of at least 0.50. Mean IoU describes how closely the correct boxes fit; it excludes missed faces and false alarms.

The model found and labelled most faces correctly. Some errors were wrong identities or extra predictions around a face. The lower score under stricter overlap requirements shows that box placement still has room to improve.

These results apply to synthetic scenes made from a small set of repeated photos. The labels were generated with a face detector. Strong scores here do not guarantee the same performance on real group photos.

Training and validation loss

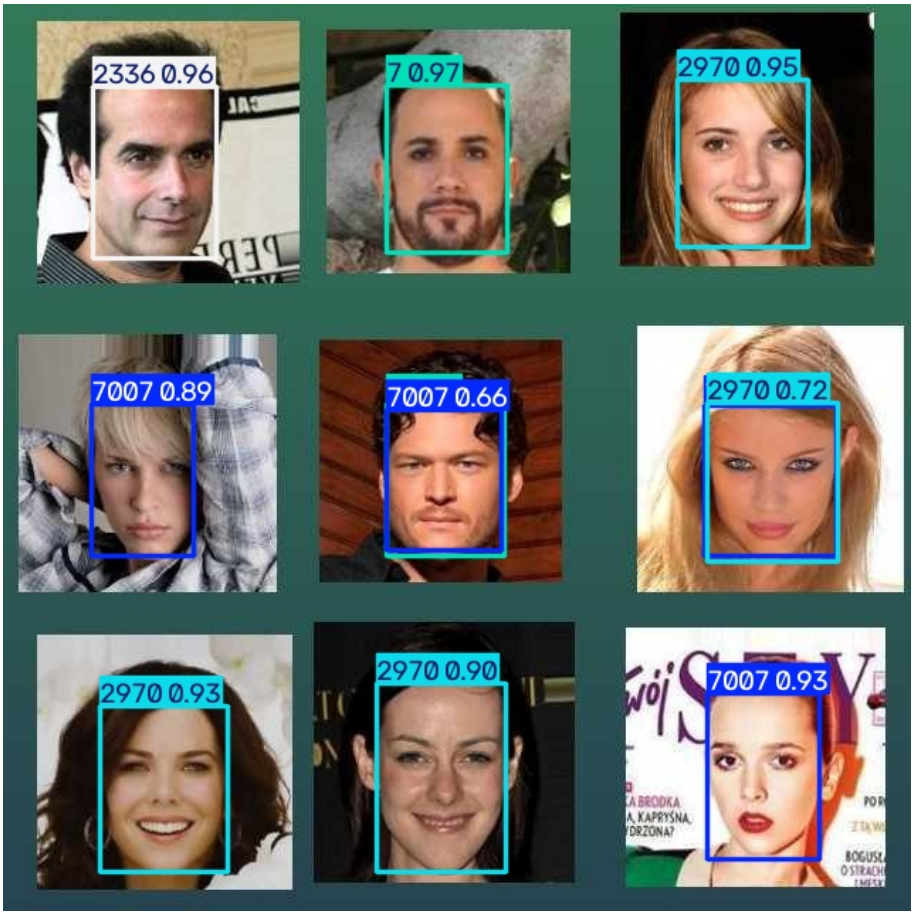
The model trained for 50 epochs. Epoch 29 had the best validation score and supplied the checkpoint tested here. The test results were not used to choose it.



Training loss fell, but validation loss stopped improving as much. The dotted line marks the selected checkpoint.

TA image check

The model placed boxes around all nine faces in the TA image. However, the middle-centre and middle-right faces each received overlapping predictions with different IDs.



No reference identity labels or boxes were supplied, so this is a visual check rather than a scored test. The model can output only its trained identities; a confident prediction can still be wrong.

Hyperparameter comparison

Dario's hyperparameter sweep results are still needed to compare training settings. This report currently covers Aditi's baseline only. If the team selects another model, its test results will need to replace these.

Detailed scores and run settings are saved in project_milestone03/results/.

Metric reference: docs.ultralytics.com/modes/val